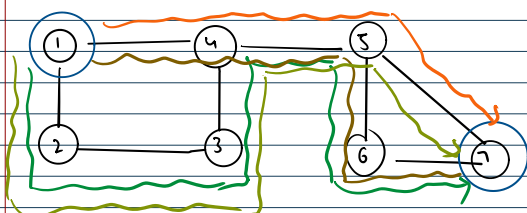
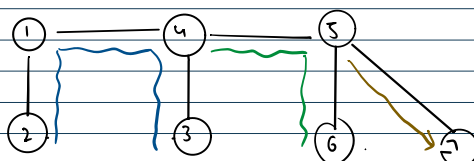
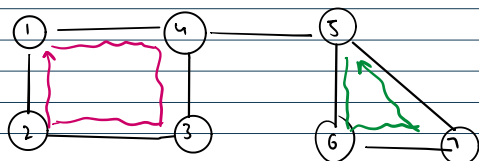
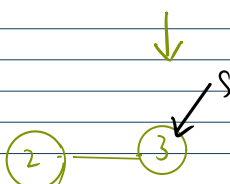
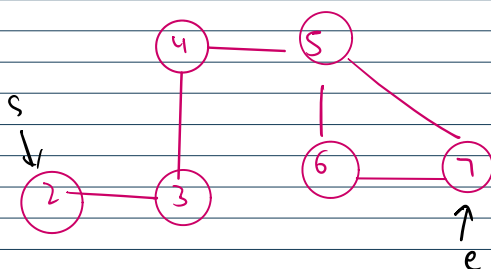
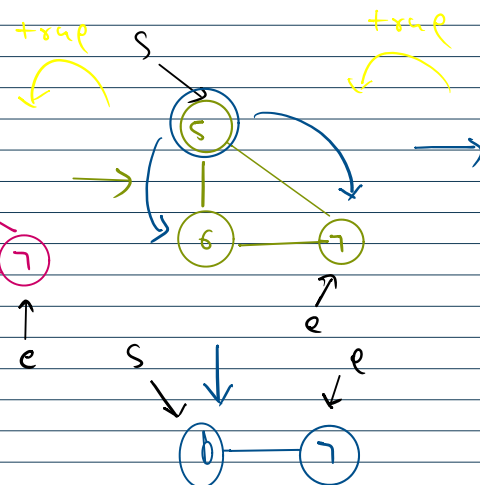
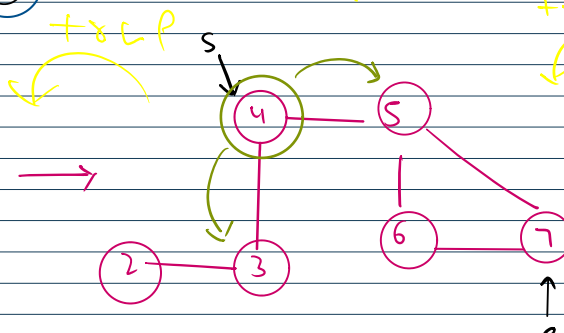
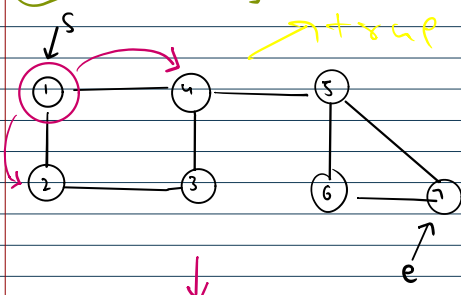


Graphs



path: 1 → 7

exists or not



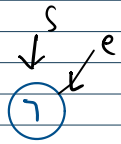
```

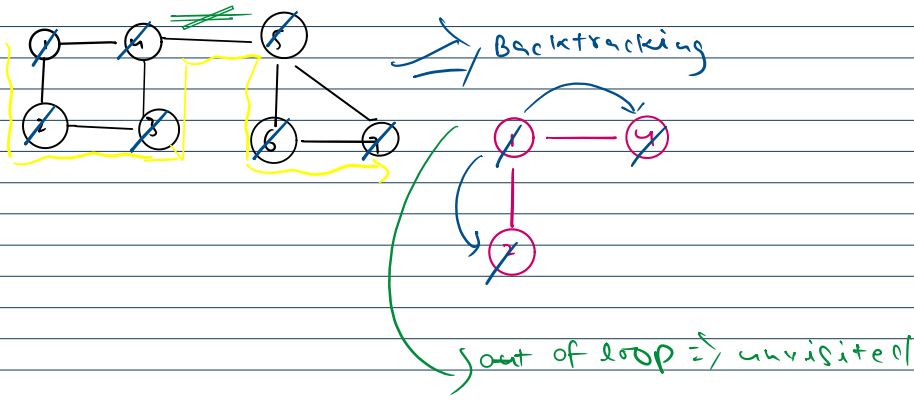
59
60 1 usage
61 public void display() {
62     for (int vtx : map.keySet()) {
63         HashMap<Integer, Integer> innerMap = map.get(vtx);
64         System.out.println(vtx + " --> " + innerMap);
65     }
66 }
67
68 1 usage
69 public boolean hasPath(int src, int des) {
70     if(src == des) {
71         return true;
72     }
73     // call hasPath for all nbrs of src
74     for(int nbrs : map.get(src).keySet()) {
75         boolean ans = hasPath(nbrs, des);
76         if(ans) {
77             return true;
78         }
79     }
80     return false;
81 }
            
```

inner map
keys of innermap

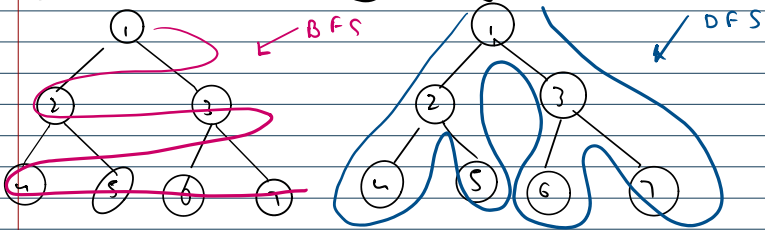


Backtracking



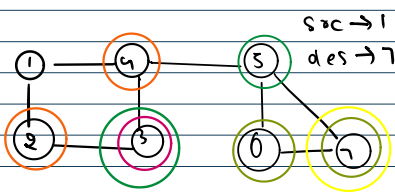


BFS (Breadth First Search)

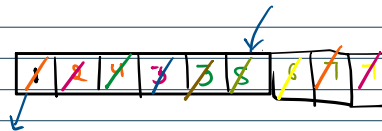


5 steps for every graph question

1. Remove
2. Ignore
3. Mark as visited
4. Self work
5. Add unvisited



BFS $\Rightarrow$ Queue $\Rightarrow$



1, 2, 4, 3
5, 6, 7
visited

1 $\Rightarrow$

2 $\Rightarrow$

4 $\Rightarrow$

3 $\Rightarrow$

3 $\Rightarrow$ Ignore

5 $\Rightarrow$

6 $\Rightarrow$

7 $\Rightarrow$ Ans.

7 $\Rightarrow$ Ignore

